Code No: 132AD **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD
B.Tech I Year II Semester Examinations, August - 2018
COMPUTER PROGRAMMING IN C
(Common to EEE, ECE, CSE, EIE, IT, ETM)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a)	Give octal number for 10001011_2.	[2]
b)	Explain different primary data types used in C language.	[3]
c)	Define binary search. How it different from linear search?	[2]
d)	Write a function that checks whether a given year is leap year or not.	[3]
e)	List any four string manipulation functions?	[2]
f)	Comment on the size of pointer to different datatypes (int *, char *, float *).	[3]
g)	What do you mean by command line arguments?	[2]
h)	Differentiate between structure and union.	[3]
i)	Discuss about ftell() function.	[2]
j)	What is meant by flushing of a file?	[3]

PART-B

(50 Marks)

2. Write a C program that reverses a number that is entered by the user by making use of do-while loop. How is this looping technique different from while loop? [10]

OR

3.a) What is an Identifier? List the rules required to form variable names in C.
b) Write a program showing the use of if else and switch statements in C? [5+5]

4. Write a program to check whether an array is ordered. If ordered print a suitable message as “Ascending” or “Descending”. Otherwise “not ordered”. [10]

OR

5. What do you mean by recursion? What conditions should be mandatory for writing a recursive function? Explain using a suitable C program. [10]

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6. A C program contains the following declaration.

static int x[8] = {10, 20, 30, 40, 50, 60, 70, 80};

a) What is the meaning of x
b) What is the meaning of (x + 2)
c) What is the value of *x
d) What is the value of (*x + 2)
e) What is the value of *(x + 2) [10]

OR

7. Write a program to copy input to output, replacing each string of one or more blanks by a single blank. [10]

8. What is bit fields? Write a program to print the binary equivalent for a given number using bit fields. [10]

OR

9. Write a program to calculate the grade of the student in a class of 60 students using structures. [10]

10. What operations can be performed on binary files? Explain. [10]

OR

11. Write a program to find the number of occurrences of a given word in a given file. [10]

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